B.Sc. BOTANY — YEAR I

MAJOR COURSE III

APPLIED BOTANY

UNIT II — Pollution, Phytoremediation and Bioremediation

Detailed Teacher-Style Notes

Definitions • Concepts • Tables • Flow diagrams • Examples • Applications • Exam points

Prepared in the same detailed format as the Major Course II notes.

UNIT II — POLLUTION AND REMEDIATION

Syllabus topics covered: pollution and pollutants; phytoremediation with five plants; bioremediation; health effects; societal efforts for pollution control; local survey of pollutants.

2.1 Pollution and pollutants

Definition — Pollution: An undesirable change in the physical, chemical or biological quality of the environment caused by the introduction or accumulation of substances or forms of energy at levels that can harm organisms, ecosystems or human welfare.

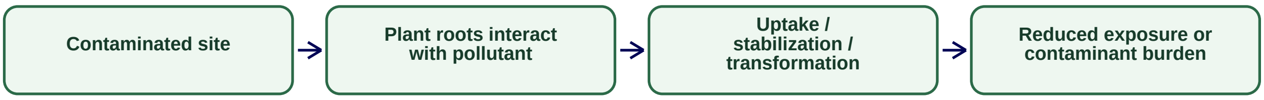
Definition — Pollutant: Any substance or form of energy that, when present in sufficient concentration, can cause adverse environmental effects.

Type	Typical sources	Main effects
Air pollution	Vehicles, industries, burning, dust	Respiratory stress, smog, plant injury
Water pollution	Sewage, industrial discharge, agricultural runoff	Disease, eutrophication, toxicity
Soil pollution	Pesticides, metals, waste dumping	Reduced soil quality and food-chain contamination
Noise pollution	Transport and machinery	Stress and hearing-related effects
Thermal pollution	Hot industrial effluents	Changes in aquatic oxygen and metabolism

2.2 Phytoremediation

Definition — Phytoremediation: The planned use of plants and their associated root-zone processes to remove, immobilize, transform or reduce contaminants in soil, water or sediments.

General phytoremediation workflow



Major mechanisms

Mechanism	Meaning
Phytoextraction	Plant uptake and accumulation of contaminants in harvestable tissues
Phytostabilization	Reduction of contaminant mobility in the environment
Rhizofiltration	Root-based removal/retention of contaminants from water
Phytodegradation	Plant-associated transformation of certain contaminants
Phytovolatilization	Conversion and release of some compounds into the atmosphere in transformed form

Five syllabus-aligned example plants and how to present them

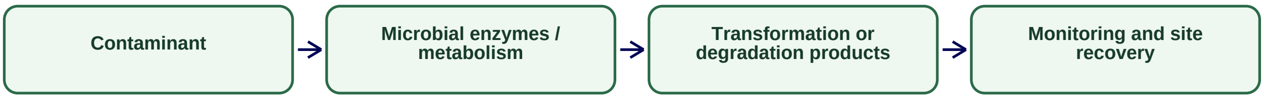
Plant	Botanical name	Teaching use in phytoremediation examples
Indian mustard	Brassica juncea	Often discussed for metal uptake/accumulation studies
Sunflower	Helianthus annuus	Used in examples of contaminant uptake and water/soil remediation research
Vetiver	Chrysopogon zizanioides	Deep roots help in stabilization and treatment systems
Water hyacinth	Eichhornia crassipes	Known for uptake from nutrient-rich/contaminated water; invasive status must also be considered
Cattail	Typha spp.	Useful in constructed wetlands and root-zone treatment systems

Important limitation: phytoremediation is not magic. It can be slow, depth-limited and dependent on plant survival, contaminant concentration and safe handling of contaminated biomass.

2.3 Bioremediation

Definition — Bioremediation: The use of living organisms, especially microorganisms and their metabolic activities, to degrade, transform, detoxify or immobilize environmental contaminants.

Basic bioremediation concept



- **In situ:** treatment at the contaminated site.
- **Ex situ:** contaminated material is removed and treated elsewhere.
- **Biostimulation:** conditions are adjusted to stimulate useful indigenous microbes.
- **Bioaugmentation:** selected microorganisms are introduced where appropriate.

2.4 Health effects of pollutants

Pollutant category	Possible health concern
Fine particulate matter	Respiratory and cardiovascular stress
Toxic gases	Irritation and reduced respiratory function depending on exposure
Contaminated water	Water-borne illness and chemical toxicity
Heavy metals	Organ-specific toxicity and long-term effects depending on metal and dose
Pesticide exposure	Acute or chronic effects depending on compound and exposure

2.5 Efforts of society for pollution control

Pollution-control hierarchy

- Use cleaner technologies and energy sources.
- Segregate and scientifically manage waste.
- Improve public transport and reduce open burning.
- Protect green spaces and water bodies.
- Promote awareness, monitoring and compliance with environmental rules.

2.6 Local survey project

Survey format: location → pollutant type → probable source → visible evidence → affected people/plants → existing control measure → suggested solution. Do not claim laboratory concentration values unless you actually measured them.

Exam focus: Unit II high-yield: definition of pollutant, classification, phytoremediation mechanisms, five plants with botanical names, bioremediation types, health effects and pollution-control hierarchy.